

Comprehensive Error Profiling of NovaSeq6000, NovaSeqX, and Sxxxx Pro Using Overlapping Paired-End Reads

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Abstract

Accurate base calling is essential for ensuring the reliability of next-generation sequencing (NGS) data. However, current approaches to error profiling often rely on reference genomes or variant-calling pipelines, which can introduce platform-specific biases or overlook systematic sequencing errors. Here, we present ErrorProfiler, a reference-free, paired-end overlap-based workflow for quantifying and characterizing base-call errors. We applied this approach to profile three high-throughput sequencing platforms - NovaSeq6000, NovaSeqX, and Sxxxx Pro - using *Escherichia coli* datasets. By analyzing mismatches within fully overlapping 150 bp read regions, ErrorProfiler effectively distinguished sequencing-specific errors from PCR or library preparation artifacts.

NovaSeqX exhibited the lowest median sequencing error rate (0.089‰), with minimal cycle-dependent drift. Sxxxx PRO followed (0.124‰), outperforming NovaSeq6000 (0.185‰). Base quality score calibration analysis revealed systematic underestimation of base-call accuracy across all platforms. Illumina's binned Q-score models (Q11/Q25/Q37; Q12/Q24/Q40) masked substantial intra-bin variability, while Sxxxx Pro's continuous Q-score system offered finer granularity but still overestimated error probabilities in key bins.

A previously unreported “upstream effect” was observed, where the empirical accuracy of a base-call was significantly influenced by the Q-score of its preceding base. This effect was most pronounced in high-Q bins and consistent across all platforms. Substitution error spectra revealed distinct platform-specific and sequence-context-dependent biases, driven by optical systems and detection chemistry.

Our findings provide a rigorous, platform-agnostic framework for sequencing error analysis and quality assessment. ErrorProfiler is open-source and compatible with diverse sequencing platforms, offering a valuable tool for improving calibration, variant calling, and downstream bioinformatics workflows.

Introduction

The advent of next-generation sequencing (NGS) has significantly reduced the cost of whole genome sequencing (WGS), with current estimates approaching \$200 per human genome [1].

This reduction has catalyzed widespread adoption of NGS across a broad spectrum of

applications, ranging from precision medicine to evolutionary and population genomics. At the core of NGS reliability lies accurate base calling, which is essential for ensuring the validity of downstream analyses, including variant detection, expression profiling, and metagenomic classification [2].

Sequencing errors can originate from multiple stages of the NGS workflow: (i) sample handling and preservation, (ii) polymerase chain reaction (PCR)-based amplification during library preparation or cluster formation on solid phase (glass plates or beads), and (iii) the sequencing process itself [3]. Artifacts introduced during sample preparation are particularly evident in formalin-fixed, paraffin-embedded (FFPE) tissues, where cytosine deamination leads to elevated C:G>T:A transitions and DNA fragmentation contributes to random base substitutions [4-6]. While some of these effects may be mitigated through strand bias analysis or increased sequencing depth, chemical degradation of nucleic acids remains a major source of error [7].

Errors introduced during PCR amplification—such as polymerase misincorporation—are non-negligible even with high-fidelity enzymes, with estimated rates of ~4.2 errors per million bases [8]. Sequencing errors, particularly those inherent to sequencing-by-synthesis (SBS) platforms, are often systematic and difficult to eliminate. Factors such as cluster phasing/pre-phasing, base-specific miscalls, and optical artifacts contribute to platform-specific error signatures. For instance, Illumina instruments frequently exhibit errors in specific sequence motifs (e.g., GGC), leading to recurrent T>G transversions, while NovaSeq systems display position-specific errors associated with patterned flow cell chemistry [9-11]. These biases can persist even after applying sophisticated base-calling and error-correction algorithms [12].

Despite the central role of base-calling accuracy in evaluating platform performance, accurately quantifying sequencing error rates and pinpointing their sources remains challenging. Benchmarking studies typically rely on well-characterized reference genomes—such as Genome in a Bottle (GIAB) samples [13], 1000 Genomes Project datasets [14], or microbial reference strains—to assess performance using variant calling pipelines. However, such reference-based approaches are often limited by the choice of alignment algorithms and variant callers. Notably, machine learning-based variant callers may be biased toward platforms used during model training, resulting in reduced generalizability [15].

Moreover, cross-platform comparisons generally require constructing matched libraries and sequencing them under tightly controlled conditions—a process that is costly and impractical for routine quality assessment. An alternative strategy proposed by Stoler et al. [10] leverages redundant base calls in overlapping regions of paired-end (PE) reads to directly detect sequencing errors without a reference genome. Because the likelihood of identical sequencing errors occurring in both reads is negligible, observed mismatches in these overlapping regions can be confidently attributed to sequencing-specific errors rather than biological variation or sample-related artifacts.

In this study, we introduce ErrorProfiler, a reference-free, modular pipeline that uses overlapping PE reads to disentangle sequencing errors from upstream sample or library preparation artifacts. Our workflow integrates base-level error quantification and visualization, enabling rapid, platform-agnostic profiling of substitution patterns and quality score accuracy. We applied ErrorProfiler to benchmark three sequencing platforms: the widely adopted NovaSeq6000, the newer NovaSeqX system, and Sxxxx Pro, a novel instrument from Sxxxx Biomed Inc. Our results not only offer the first detailed characterization of NovaSeqX and Sxxxx Pro error profiles but also highlight systematic differences in quality score calibration, substitution biases, and the influence of upstream sequence context on base-call accuracy.

Results

Sequencing Error Rate and Its Proportion in Total Base Call Error

To evaluate the contribution of sequencing-specific errors to overall base-calling inaccuracies, we first analyzed the cycle-dependent progression of total base-call error rates across three platforms: NovaSeq6000, NovaSeqX, and Sxxxx Pro (Figure 1A). Sequencing error rates were then estimated by quantifying mismatches within the overlapping regions of paired-end reads (Figure 1B) and its proportion in total base-call error (Figure 1C). As this method is constrained by read length, it enables accurate estimation only within the overlapping 150 base pairs per read.

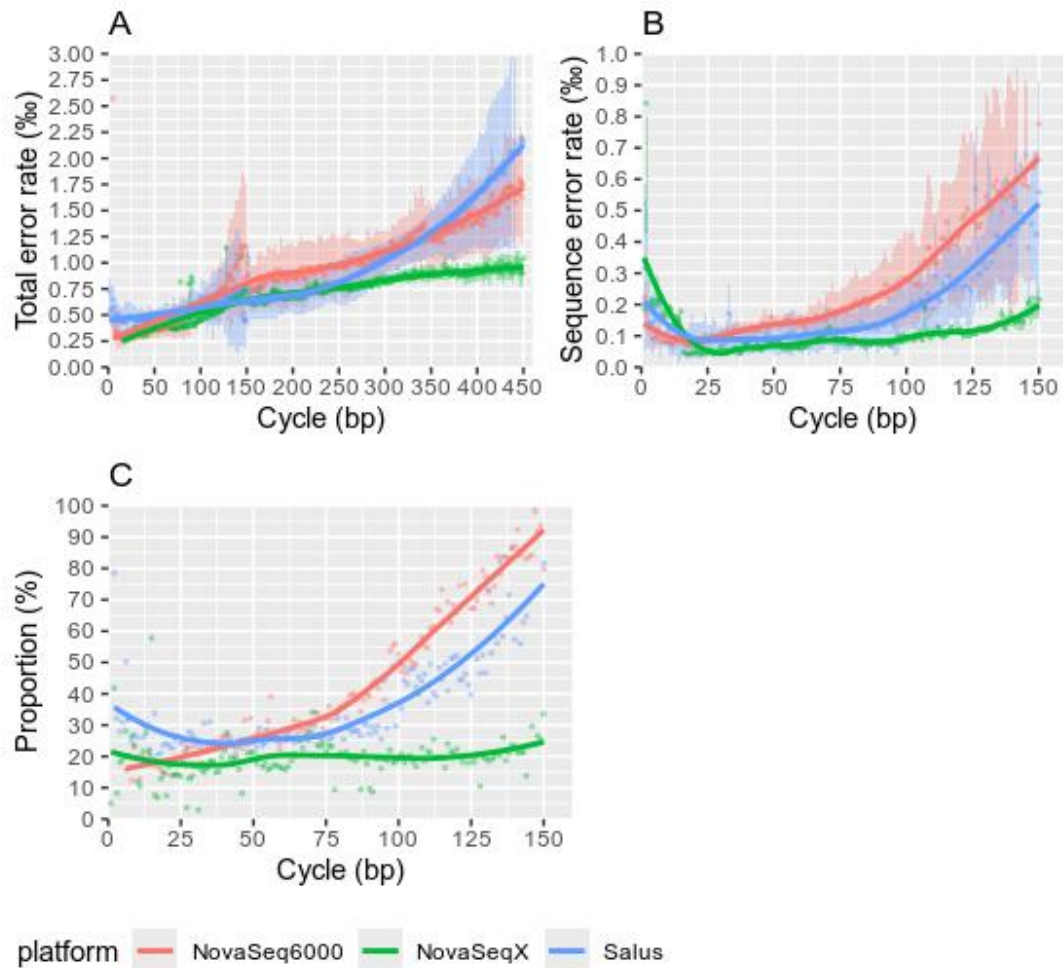


Figure 1 Sequencing error rate dynamics and their contribution to total base-call errors across three platforms. (A) Total base-call error rate plotted as a function of sequencing cycles for NovaSeq6000, NovaSeqX, and Sxxxx PRO platforms. NovaSeqX displays the lowest absolute error rate and the slowest accumulation over cycles, while NovaSeq6000 shows rapid error escalation. Sxxxx Pro exhibits a biphasic pattern, with a slow increase followed by accelerated growth beyond 200 cycles. (B) Sequencing error rates estimated from mismatches in overlapping paired-end reads. NovaSeqX maintains stable error rates throughout, whereas NovaSeq6000 and Sxxxx Pro show pronounced increases after the initial cycles. (C) Contribution of sequencing errors to total base-call errors across read positions. NovaSeq6000 transitions rapidly from PCR-dominated to sequencing-dominated errors, reaching ~90% sequencing-derived error by cycle 150. In contrast, NovaSeqX maintains a stable ~20% contribution, while Sxxxx PRO exhibits a biphasic shift from stable to increasing sequencing error proportions beyond cycle 75. Median values reflect a distinct hierarchy: NovaSeqX shows the lowest sequencing error proportion, followed by Sxxxx PRO and NovaSeq6000. These profiles suggest varying amplification-to-sequencing error equilibria across platforms.

We next examined the sequencing-specific error rate and its relative contribution to total base-call errors (Figures 1B–C). Following initial 10 cycles instability, sequencing error rates diverged notably among platforms. NovaSeq6000 demonstrated a steady increase from

0.10‰ at cycle 10 to 0.675‰ at cycle 150. In contrast, NovaSeqX maintained a nearly flat profile from cycle 10 to 150 (0.05‰ to 0.20‰), indicating strong base-calling consistency across read length. Sxxxx Pro showed an intermediate trajectory: its initial error rate (0.08‰) was comparable to NovaSeq6000, but its accumulation was more gradual, reaching 0.55‰ at cycle 150. Median sequencing error rates across the 150-cycle window ranked as follows: NovaSeqX (0.089‰) < Sxxxx Pro (0.124‰) < NovaSeq6000 (0.185‰).

Figure 1C depicts the evolving proportion of sequencing errors relative to total base-call errors across cycles. For NovaSeq6000 and Sxxxx Pro, sequencing errors became increasingly dominant toward later cycles. NovaSeq6000 showed the steepest increase in sequencing-derived error contribution, rising from 15% at cycle 1 to 90% by cycle 150. Sxxxx Pro followed a biphasic trend, with sequencing errors accounting for ~25% before cycle 75 and escalating to 75% by cycle 150. In contrast, NovaSeqX maintained a relatively constant sequencing error contribution of ~20% across all cycles. These trends indicate distinct balances between amplification- and sequencing-derived error sources among platforms. Median contributions of sequencing errors to total base-call errors were lowest in NovaSeqX (20.0%), followed by Sxxxx Pro (29.0%), and highest in NovaSeq6000 (34.0%).

Accuracy of base quality score and the Upstream Effect

To assess the accuracy of base quality scores reported by each sequencing platform, we compared FASTQ-reported Q-scores against empirically derived Q-scores calculated from base concordance in overlapping paired-end reads (Figure 2). Across all three platforms, over 95% of data points were positioned above the diagonal in the Q-Q plot, indicating that reported Q-scores systematically underestimated true base-call accuracy.

Illumina platforms (NovaSeq6000 and NovaSeqX) employed a binned quality scoring system, assigning base calls to three discrete Q-score levels. This binning approach substantially reduced the resolution of quality assessment and obscured fine-scale differences in base-call accuracy. NovaSeqX categorized reads into Q12 (low), Q24 (medium), and Q40 (high), with 99.2% of base calls falling within the Q40 bin. However, the empirical accuracy of bases within this bin was substantially higher, corresponding to an average Q-score of 55.7—a 39.3% inflation relative to the platform's error model. Similarly, NovaSeq6000 utilized Q11, Q25, and Q37 bins, with 96.9% of bases in the Q37 category, which corresponded to a true accuracy equivalent to Q53.7 (45.1% inflation). These findings underscore the limited granularity of binned scoring systems in reflecting true base-call reliability. By contrast, Sxxxx Pro adopted a continuous Q-score system ranging from 1 to 39, allowing for more nuanced representation of base-call confidence. Most base calls (92.9%) were assigned scores greater than Q30, consistent with high nominal quality. Nonetheless, benchmarking revealed a polymodal distribution of Q-score inflation. The most frequent scores (Q32 at 12.6% and Q37 at 12.9%) significantly underestimated actual accuracy. Q32, for instance, corresponded to an empirical score of 51 (59.3% inflation), while Q37 mapped to Q53.7 (45.1% inflation). Moreover, only 52.3% of bases achieved empirical Q-scores above Q53 (corresponding to >99.95% accuracy), the lowest among the three platforms. While all platforms overestimated error probabilities to some degree, Sxxxx Pro's continuous scoring model

provided finer granularity for distinguishing high-accuracy base calls, with nominal Q30 bases approximating Q46.4 in empirical quality.

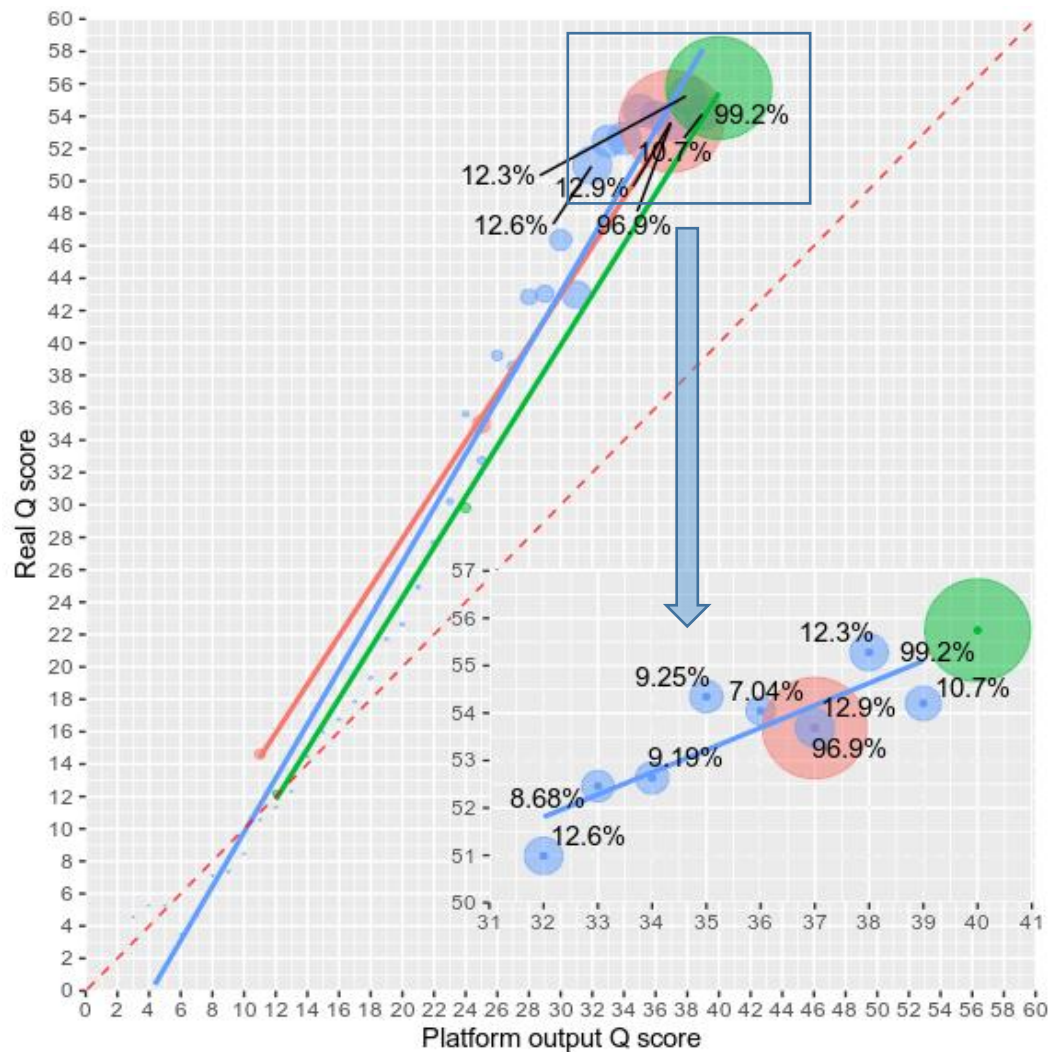


Figure 2 Comparison of reported versus empirical base quality scores across sequencing platforms. Q-Q plots depict the relationship between platform-reported Q-scores (x-axis) and empirically derived Q-scores based on concordance in overlapping paired-end reads (y-axis). The red dashed diagonal line represents perfect concordance. Data points above the line indicate underestimation of base-call accuracy. Circle size corresponds to the proportion of total base calls at each Q-score, and color denotes the sequencing platform (green: NovaSeqX; red: NovaSeq6000; blue: Sxxxx Pro). Insets magnify the high-Q-score region, highlighting discrepancies between reported and empirical accuracy. Notably, Illumina platforms (NovaSeq6000 and NovaSeqX) employ discrete Q-score binning, resulting in inflated quality estimates—particularly at Q37 and Q40. In contrast, Sxxxx Pro's continuous Q-score system enables finer granularity but still underestimates accuracy in its most frequent bins (Q32 and Q37). Percentages indicate the relative frequency of each score bin.

In addition to global deviations between reported and actual base-call accuracy, we identified a previously uncharacterized phenomenon: the upstream effect, where the quality score of the preceding base influences the empirical accuracy of the current base, even when its reported

Q-score remains constant (Figure 3). This context dependency persisted across all platforms and was particularly pronounced for high-quality base calls.

To quantify the magnitude of this effect, we performed linear regression between upstream Q-score and empirical accuracy of the current base, stratified by reported Q-score (Table 1). Regression coefficients were consistently larger for high-quality bins (NovaSeq6000 Q37: 0.451; NovaSeqX Q40: 0.687; Sxxxx Pro Q37: 1.211), indicating that these bases are more susceptible to upstream signal quality than their lower-quality counterparts. For instance, on Sxxxx Pro, a base reported as Q37 exhibited an empirical Q-score of 41 when preceded by a Q27 base, but increased to Q53 when the upstream base was also Q37—a difference of 12 Q units. In contrast, low- and mid-quality bases exhibited weaker dependencies (e.g., Sxxxx Pro Q12: 0.071).

While Illumina platforms also showed this effect, their binned Q-score structure limited its resolution. For example, NovaSeq6000 Q37 base calls had empirical accuracies ranging from Q43.7 to Q55.5 depending on upstream base quality. On NovaSeqX, the empirical accuracy of Q40 calls decreased to Q38 when preceded by a Q12 base. These findings indicate that base-calling errors are not independent across positions, challenging assumptions embedded in many quality recalibration tools (e.g., GATK BQSR), which typically exclude upstream base quality as a covariate.

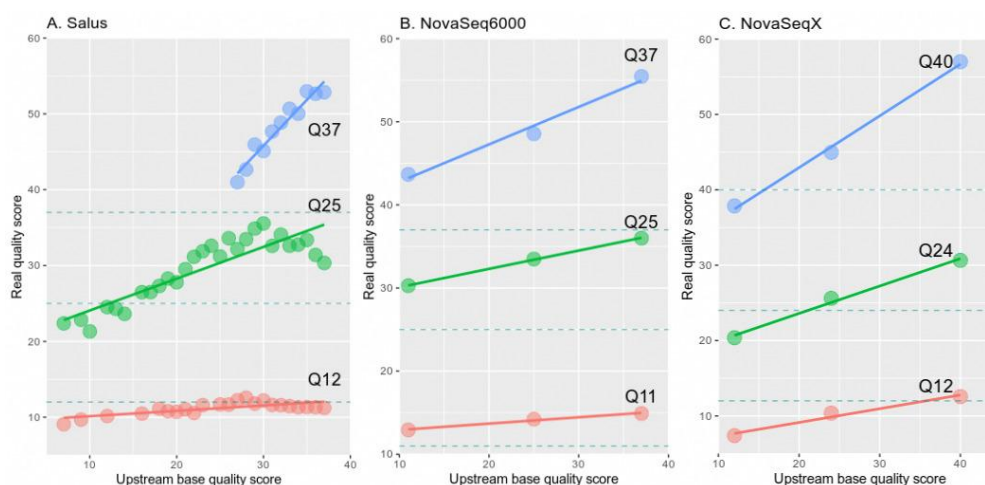


Figure 3 Upstream base Q-score influences empirical base-call accuracy across three sequencing platforms. Scatter plots show the relationship between the upstream base's reported Q-score (x-axis) and the empirical Q-score of the current base (y-axis), stratified by nominal Q-score categories (colored lines) across three sequencing platforms: (A) Sxxxx Pro, (B) NovaSeq6000, and (C) NovaSeqX. Each point represents a specific upstream Q-score bin and fitted regression lines are overlaid for each quality tier. Blue, green, and red lines represent high, medium, and low reported Q-score groups, respectively (e.g., Q37/Q40, Q25/Q24, Q12/Q11). Dashed horizontal lines indicating the nominal Q-score across each bin was draw as a reference. A strong positive correlation is evident for high-Q-score bases, particularly on Sxxxx Pro (Q37) and NovaSeqX (Q40), suggesting that base-call accuracy is strongly dependent on upstream signal quality. In contrast, low-quality bins (Q11/Q12) exhibit little to no dependence, indicating limited sensitivity to context. These findings collectively define a platform-wide "upstream effect" that is most pronounced for nominally high-quality base calls. Please

note that for a clear view and comparison we display only three nominal Q-score categories for Sxxxx Pro, even though it uses continuous Q-score system and the effect exists almost all bases.

Table 1 Linear regression coefficients quantifying the influence of upstream base Q-score on empirical base-call accuracy across three sequencing platforms.

Platform	Base quality score batch	Coefficients	Adjusted R ²	P-value
Sxxxx Pro	Q12	0.071	0.5378	4.135e-05
	Q25	0.419	0.7654	6.946e-10
	Q37	1.211	0.9327	8.81e-07
NovaSeq6000	Q11	0.076	0.9655	0.0839
	Q25	0.220	0.9987	0.01611
	Q37	0.451	0.9592	0.09129
NovaSeqX	Q12	0.182	0.9434	0.1076
	Q24	0.364	0.9831	0.05866
	Q40	0.687	0.9915	0.04148

Substitution Bias and Context Dependency in Sequencing Errors

To investigate platform-specific substitution patterns, we quantified the proportion of each of the 12 possible single-nucleotide substitution types across NovaSeq6000, NovaSeqX, and Sxxxx Pro platforms (Figure 4). If substitution errors were randomly distributed, each of the 12 types would comprise approximately 8.3% (1/12) of total substitutions. However, all three platforms exhibited distinct and non-random error spectra, reflective of platform-specific optical chemistries and base-calling mechanisms.

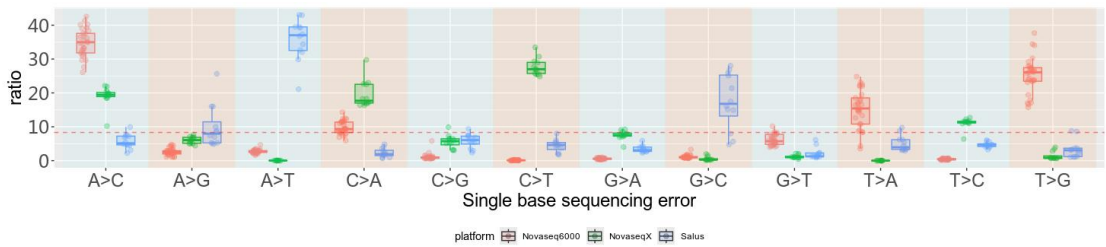


Figure 4 Platform-specific biases in single-nucleotide substitution errors. Box plots show the distribution of substitution error proportions across 12 canonical single-base changes (e.g., A>C, T>G) for NovaSeq6000 (red), NovaSeqX (blue), and Sxxxx PRO (green). Each substitution type is normalized within each sequencer such that all 12 types sum to 100% per sample. The dashed horizontal line marks the expected value for uniform substitution distribution (8.3%). NovaSeq6000 exhibits dominant A>C and T>G errors, while NovaSeqX shows enrichment for C>A, C>T, and A>G substitutions. Sxxxx PRO displays a distinct error spectrum with A>T and G>C substitutions predominating. These differences reflect underlying detection chemistries and base-calling models specific to each platform.

NovaSeq6000 displayed a pronounced bias toward A>C (median: 36.1%) and T>G (25.1%) substitutions, with a modest enrichment of T>A errors (15.0%). These patterns align with prior studies reporting spectral cross-talk in the two-color (red/green) detection system of Illumina instruments [9]. Specifically, adenine is detected via a combined red and green signal, while thymine and cytosine are each represented by individual color channels (green and red, respectively), and guanine by the absence of signal (“dark base”). The overrepresentation of A>C and T>G errors may result from imperfect detection or deconvolution of green channel signals, while the modest T>A signal could reflect cross-channel leakage from red to green (Table 2).

Table 2 Color mapping schematic for NovaSeq6000

Observed Error	Original color signal)	Erroneous color signal
A>C (Major)	Red + Green	Red
T>G (Major)	Green	No Signal
T>A (Minor)	Green	Red + Green

In contrast, NovaSeqX—which uses a blue/green two-channel detection system—exhibited a different error spectrum dominated by C>A, C>T, and A>C substitutions. These shifts suggest that NovaSeqX’s updated chemistry (XLEAP-SBS) and optical design mitigate traditional red-green cross-talk but introduce new forms of channel imbalance (Table 3). Sxxxx Pro, which employs a four-channel system, presented a distinct error profile, with A>T and G>C substitutions comprising the majority of errors. These may result from partial spectral overlap or lower signal intensity associated with the dyes used to label adenine and guanine.

Table 3 Color mapping schematic for NovaSeqX

Observed Error	Original Base (Color Signal)	Erroneous Base (Color Signal)
A>C (Major)	Blue	Blue + Green
C>A (Major)	Blue + Green	Blue
C>T (Major)	Blue + Green	Green

To further explore sequence-context dependencies, we extended the substitution analysis to include the upstream base (i.e., dinucleotide context), revealing additional platform-specific biases (Figure 5). On NovaSeq6000, the most frequent substitution motif was AA>AC (median: 18%), and both CT>CA and TT>TA substitutions were significantly more common than context-matched alternatives (e.g., AT>AA, GT>GA; Mann–Whitney U test, p -value 5.1×10^{-2}). These patterns are consistent with phasing effects—where incomplete extension of DNA strands during SBS cycles results in signal mixing between adjacent bases. Notably, however, C>A substitutions were not strongly enriched on NovaSeq6000, suggesting that this artifact may be less prevalent than previously reported in studies relying solely on alignment-based error detection.

On NovaSeqX, context-dependent substitution biases were particularly evident for C>A and C>T events. For example, upstream adenine (A) or cytosine (C) strongly increased the likelihood of these substitutions, highlighting the impact of neighboring base identity on polymerase behavior or dye interpretation. Sxxxx Pro displayed strong A>T context bias when the upstream base was also A or G, making AA>AT and GA>GT the most frequently observed dinucleotide errors. These findings further reinforce the notion that systematic substitution errors are not only base-type specific but also strongly modulated by local sequence context.

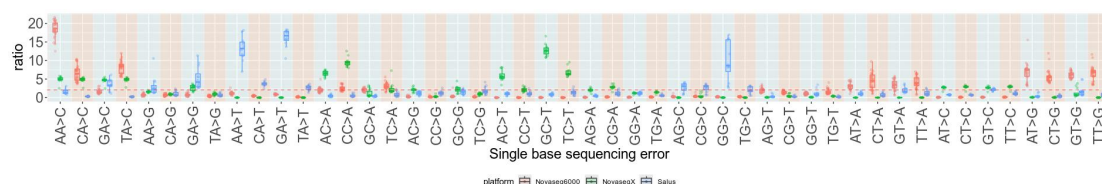


Figure 5 Sequence context influences substitution bias at the dinucleotide level. Box plots illustrate the substitution error ratio for each dinucleotide context, where the upstream base and the substituted base jointly define the motif (e.g., AA>C, CT>A). Error ratios are grouped by platform: NovaSeq6000 (red), NovaSeqX (blue), and Sxxxx PRO (green). NovaSeq6000 shows strong enrichment of AA>AC and TT>TA substitutions, suggestive of phasing-related signal interference. NovaSeqX exhibits elevated C>A and C>T errors when preceded by A or C. Sxxxx Pro is characterized by context-sensitive A>T and G>T errors, particularly in AA and GA contexts. These findings underscore the importance of local sequence context in shaping platform-specific error profiles.

Collectively, these results illustrate that substitution biases are influenced by platform-specific chemistry, fluorescence detection systems, and sequence context. Such patterns must be carefully considered in downstream applications such as variant calling, where unmodeled systematic errors can lead to spurious variant calls in low-frequency or low-complexity regions.

Discussion

Accurately evaluating the base-calling performance of next-generation sequencing (NGS) platforms remains challenging due to the multiple sources of noise arising from sample handling, library preparation, and sequencing chemistry. Existing studies primarily benchmarked sequencing accuracy by comparing results against reference genomes from well-characterized standards such as GIAB human samples or ATCC microbial strains. In such cases, single-nucleotide variants (SNVs) or short InDel are identified either through variant-calling software (e.g., GATK Mutect2, Strelka2, CASAVA) applied to aligned reads [9], [16-17], or by directly parsing CIGAR strings and MD tags from alignment files [18-22].

However, the former approach is inherently dependent on the calibration and filtering logic of the variant caller, while the latter relies on the assumption that the reference genome perfectly represents the sequenced template—an assumption that rarely holds true in

practice.

To overcome these limitations, we adopted an alternative strategy originally proposed by Stoler et al. [10], based on assessing consistency in overlapping regions of paired-end (PE) reads. This method circumvents reliance on external standards or variant-calling pipelines by inferring errors directly from base-level discordance between read pairs covering the same genomic loci. In this study, we extended this principle into a robust, modular workflow named ErrorProfiler, and applied it to profile three NGS platforms: NovaSeq6000, NovaSeqX, and Sxxxx Pro. Notably, this is the first comprehensive characterization of NovaSeqX and Sxxxx Pro error profiles using a reference-free design.

To reduce high-cycle sequencing artifacts, we restricted our analysis to 150 bp fully overlapping read regions. Under these constraints, NovaSeqX exhibited the lowest median sequencing error rate (0.089‰) and the most stable error growth over read cycles (0.05–0.2‰). Sxxxx PRO followed, with a modest error rate (0.124‰) but more pronounced cycle-dependent variation (0.08–0.55‰). NovaSeq6000 recorded the highest error rate (0.185‰) and the fastest accumulation of sequencing errors. These rates are substantially lower than those reported in prior studies (e.g., 1.09‰ for NovaSeq6000 in [10]), likely due to stricter filtering of overlap lengths and improved public dataset quality.

We next assessed the calibration accuracy of platform-reported base quality scores (Q-scores) using Q-Q plots. Across all three platforms, >95% of data points fell above the diagonal, indicating systematic underestimation of base-calling accuracy. This effect was especially pronounced for Illumina platforms, which employ a three-bin Q-score model (e.g., Q11/Q25/Q37 for NovaSeq6000; Q12/Q24/Q40 for NovaSeqX). These coarse bins masked substantial within-bin variability; for example, the Q37 bin on NovaSeq6000 empirically corresponded to Q53.7, representing a 45.1% inflation. In contrast, Sxxxx Pro adopted a continuous Q-score scale, offering more granularity. Despite this, we observed overconfidence in the most frequent scores (e.g., Q32 and Q37), where empirical accuracy was significantly higher than reported. Only 52.3% of Sxxxx Pro bases achieved empirical $Q > 53$, indicating room for calibration refinement.

A key novel finding from our study is the “upstream effect”—the observation that the empirical accuracy of a base is strongly influenced by the Q-score of the immediately preceding base. This directional dependency was most pronounced for high-quality base calls and was consistently observed across all three platforms. For instance, on Sxxxx PRO, a Q37 base had an empirical accuracy of Q41 when preceded by a Q27 base, but increased to Q53 when preceded by another Q37 base. Linear regression confirmed that this effect was stronger for high-Q bins than for lower-Q bins, with regression coefficients reaching 1.211 for Sxxxx PRO Q37 bases. Illumina platforms exhibited the same trend, although the binning structure limited resolution. This context-dependent behavior is not accounted for in widely used base quality score recalibration tools such as GATK’s BQSR [23], which models Q-score using

factors like cycle number, read direction, and dinucleotide sequence context, but omits upstream base quality as a covariate. We propose that incorporating upstream base quality into recalibration models could improve error modeling in downstream variant calling and consensus sequencing applications.

Substitution bias analysis further revealed platform-specific error signatures. NovaSeq6000 showed strong preferences for A>C and T>G substitutions, consistent with red-green fluorescence cross-talk and detection imbalances previously reported in Illumina platforms [10]. NovaSeqX, which employs blue-green optical chemistry (XLEAP-SBS), showed a distinct profile dominated by C>A, C>T, and A>C substitutions - like those observed in NextSeq500 [24] and MiSeq [21]. These patterns likely reflect not only optical channel differences but also polymerase behavior and phasing/prephasing effects. Sxxxx Pro exhibited a unique substitution pattern with dominant A>T and G>C biases, possibly arising from chemiluminescent detection chemistry and altered spectral overlap of nucleotide labels.

Dinucleotide-level context analysis (i.e., upstream-dependent substitutions) provided further evidence of sequence-context effects. For example, NovaSeq6000 showed strong enrichment for AA>AC and TT>TA transitions, which are consistent with phasing artifacts caused by incomplete extension or misalignment between red and green signal channels [10], [25].

Similarly, NovaSeqX displayed elevated C>T and C>A rates when the upstream base was A or C, while Sxxxx Pro showed a marked increase in A>T substitutions when preceded by A or G (e.g., AA>AT, GA>GT). These trends highlight that substitution patterns are not only platform-dependent, but also sequence-context dependent, and should be incorporated into platform-specific error models.

In conclusion, this study introduces ErrorProfiler, a reference-free pipeline for characterizing sequencing error profiles based on overlapping paired-end reads. Through comprehensive profiling of three distinct sequencing platforms, we demonstrate that NovaSeqX offers the highest base-calling accuracy, with minimal sequencing-driven error accumulation. Sxxxx PRO, despite being a novel system, surpasses NovaSeq6000 in both error rate and quality score fidelity, making it a promising candidate for high-accuracy sequencing applications. Our results also reveal systematic base-calling dependencies—both in terms of local sequence context and upstream Q-score effects—that have implications for recalibration algorithms and variant-calling precision. These findings emphasize the need for platform-specific, context-aware error modeling to support accurate and reproducible genomics.

Materials and Methods

Sample Collection and Selection Criteria

To evaluate sequencing errors in publicly available Illumina platforms, we queried the NCBI

Sequence Read Archive (SRA) using the keywords: "escherichia coli"[Organism] AND "illumina"[Platform] AND "filetype fastq"[Properties]. As of February 2025, this search yielded 297,831 entries. From these, we retained only entries meeting all the following criteria:

- 1, Total base count ≥ 5 Gb
- 2, Public release date $\geq 2023-01-01$
- 3, Library layout specified as "PAIRED"
- 4, Average read length of 300 or 302 bp
- 5, Library strategy defined as 'AMPLICON', or 'RNA-Seq', or 'WGS'
- 6, Library selection method excluding 'size fractionation' or 'Inverse rRNA'
- 7, Data submitted via SRA submission system
- 8, Data not submitted by Sxxxx Biomed Inc.

Final platform classification was manually curated using the "Model" field of each entry. This yielded 34 high-quality datasets from NovaSeq6000 and 12 datasets from NovaSeqX instruments (Supplementary Table 1).

For Sxxxx Pro, we generated a dedicated dataset using *Escherichia coli* strain ATCC 8739, sourced from the Guangdong Microbial Culture Collection Center (GDMCC). Cultures were grown in-house for ~13 hours (~40 generations). DNA extraction was performed using the Bacterial Genomic DNA Extraction Kit (TIANGEN, DP302-02). Libraries were constructed using the VAHTS Universal Plus DNA Library Prep Kit for Illumina V2 (Vazyme), involving fragmentation, end-repair, A-tailing, dual-end adapter ligation, and two cycles of PCR amplification. Libraries (20–40 ng/ μ L, ~300 bp in length) were quality-checked using the 1 $\times$ dsDNA HS Assay Kit and LabChip GX Touch analyzer.

Libraries were divided into 20 technical replicates and shipped to Sxxxx Biomed Inc. for sequencing using PE150 mode with dual barcoding. Each batch was sequenced by a different technician and instrument to minimize batch effects. Final sequencing runs are available at NCBI BioProject PRJNA1277372 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1277372>)

Data Processing and Alignment

For NovaSeq datasets, .sra files were retrieved using prefetch and converted to paired-end FASTQ files using fasterq-dump. All raw reads were quality-filtered using fastp [26] with parameters: --qualified_quality_phred 25 --unqualified_percent_limit 10 --length_required 50

Filtered reads were aligned using BWA-MEM2 [27] to the *E. coli* K-12 MG1655 reference genome (RefSeq: NC_000913) for Illumina datasets and to *E. coli* ATCC 8739 (RefSeq: NC_010468) for Sxxxx Pro data. Alignment files were processed using sambamba [28] for sorting, indexing, compression, and PCR duplicate removal.

ErrorProfiler: Overlap-Based Error Profiling Pipeline

All BAM files were analyzed using the ErrorProfiler pipeline (available at

<https://github.com/yaotianran/ErrorProfiler>). The pipeline performs the following:

1, Overlap Calculation: Determines the length of overlap between read pairs. Only those with overlap ≥ 148 bp were retained. Although 150 bp full-overlaps are ideal, we relaxed this to 148 bp due to data limitations.

2, Empirical Base Assignment: Bases within overlapping regions were assigned as “correct base” according to the following rules:

- a) If the two reads agree, their shared base is assumed correct.
- b) If they disagree and one matches the reference genome, that base is assigned as correct.
- c) If both differ from the reference, the base with a quality ≥ 25 and at least 10 points higher than the alternative is correct.
- d) If unresolved, the base type with the highest summed quality score at that position (min $4\times$ coverage) is correct.
- e) If ambiguity remains, the base is marked “not available” and excluded from downstream analysis.

3, Feature Binning and Output: Base calls were annotated with read order, cycle position, called base, reported and empirical Q-scores, and context bases. Results were aggregated and summarized for error rate, substitution profile, and Q-score accuracy.

4, Visualization and Statistics: Implemented in R and ggplot2 [29]. Downstream plots include Q-Q plots, substitution bias bar plots, dinucleotide context effects, and regression modeling of the upstream effect.

Software and Code Availability

All analysis scripts and the full ErrorProfiler pipeline are publicly available under an open-source license at: <https://github.com/yaotianran/ErrorProfiler>

References

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